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Multi-stage bit line pre-charge

Abstract

Systems and method are provided for a memory circuit that includes a bit cell responsive to a bit line signal line and a bit line bar signal line configured to store a bit of data. A pre-charge circuit is configured to charge one of the bit line and bit line bar signal lines prior to a read operation, where the pre-charge circuit includes a first pre-charge component and a second pre-charge component, the first and second pre-charge components being individually controllable for charging the bit line and bit line bar signal lines.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 17/408,567, filed Aug. 23, 2021, which is a continuation application of U.S. application Ser. No. 16/785,875, filed Feb. 10, 2020, now U.S. Pat. No. 11,100,964, issued Aug. 24, 2021, each of which is incorporated herein by reference in their entirety.

BACKGROUND

(1) A bit cell of a memory (e.g., an SRAM device) may take the form of two cross coupled inverters, which act as latch storage elements, and two switches connecting the two inverters to complementary bit lines to communicate data into or out of the bit cell. The switches (e.g., NMOS pass transistors) are controlled by a word line. When the switches are off, the bit cell keeps one of its two possible steady states. To write data into a bit cell, the value to be written and its complement are placed onto the bit lines and the word line is raised simultaneously. To read a value from the bit cell, both bit lines are pre-charged high, and the word line is raised. The bit line relative to the bit cell node that contains a zero value begins discharging, providing differing signals that can be sensed and output from the memory.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a block diagram depicting a memory that includes a multi-stage pre-charge circuit according to an exemplary embodiment.
- (3) FIG. 2 is a diagram depicting a bit cell and a multi-stage pre-charge circuit in accordance with embodiments.
- (4) FIG. 3 provides timing diagrams associated with exemplary operations in embodiments of the disclosure.
- (5) FIG. 4 is a block diagram depicting a pre-charge module for controlling a first pre-charge component and a second pre-charge component of a pre-charge circuit in accordance with embodiments.
- (6) FIG. 5 depicts example pre-charge components in accordance with embodiments.
- (7) FIG. 6 is a diagram depicting another example pre-charge circuit in accordance with an embodiment.
- (8) FIG. 7 is a diagram depicting operation of control signals to limit a magnitude of word line coupling in accordance with embodiments.
- (9) FIG. 8 is a diagram depicting a memory circuit where a first pre-charge component also serves as a second pre-charge component for another memory bank in accordance with embodiments.
- (10) FIG. 9 is a flow diagram depicting a method of controlling a multi-stage pre-charge circuit for charging one of a bit line and bit line bar signal line associated with a bit cell prior to a read

operation in accordance with embodiments.

DETAILED DESCRIPTION

(11) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(12) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(13) As noted above, in certain memories, read operations are initiated by pre-charging two bit lines (e.g., BL, BLB) to which a bit cell is connected to a high level. When a word line (e.g., WL) of the bit cell is activated, the bit cell pulls a particular one of the bit lines low. A sense amplifier senses the difference between the two bit lines, and outputs a corresponding data value accordingly. The bit lines are then pre-charged again to facilitate a next read operation. To facilitate high speed operation, it may be desirable to start a next cycle of pre-charging promptly following the conclusion of the previous read (e.g., start pre-charging immediately after the WL goes low). In some instances, such as based on fabrication process variations, the desired timing may not be realized in practice. Untimely signals (e.g., a pre-charge control signal being activated before the word line goes low) can result in anomalous operation. For example, coupling between a charging bit line and a word line can result in disturbance of the bit cell state, potentially changing a bit cell value during a read operation.

(14) FIG. 1 is a block diagram depicting a memory that includes a multi-stage pre-charge circuit according to an exemplary embodiment. A memory **100** includes a two dimensional array of SRAM bit cells. Each row **102** of bit cells is connected to a word line WL[0] . . . WL[n] that provides control to the bit cells of a data word. Each column **104** of bit cells is associated with an input/output (IO) circuit **106** that includes a multi-stage pre-charge circuit **108**. The multi-stage pre-charge circuits **108** receive first and second control signals from a pre-charge control circuit **110** that manages pre-charge operations in the pre-charge circuits **108**.

(15) FIG. 2 is a diagram depicting a bit cell and a multi-stage pre-charge circuit in accordance with embodiments. As described above, a bit cell **202** comprises a pair of cross coupled inverters formed via two NMOS and two PMOS transistors. Data is stored in the bit cell **202** by applying different signal levels to the T and C nodes of the bit cells during a write operation via BL and BLB by controlling of switch transistors **204**, **206** using the WL signal. Upon deassertion of the WL signal, the applied differential signals will be maintained at the T and C nodes, indicating a stored data value (e.g., 1 when T is high and C is low and 0 when T is low and C is high, or vice versa).

(16) During a read operation both BL and BLB are pre-charged high. When the read operation is initiated by asserting WL, the T or C node having the low state will pull its corresponding bit line low via the ground node (i.e., if T is low, BL will be pulled low upon assertion of WL; if C is low, BLB will be pulled low upon assertion of WL). A sense amplifier senses the polarity of the

difference between BL and BLB and outputs a corresponding value of the bit read from the bit cell **202**. The low bit line must then be pre-charged again (re-charged) before another read operation can occur. Thus the pre-charge time limits the speed at which data can be read from the memory. (17) An example two-stage pre-charge circuit is depicted at **208**. The pre-charge circuit includes a first pre-charge component **210**, labeled weak pre-charge, and a second pre-charge component **212**. The first **210** and second **212** pre-charge components are individually controllable for charging the BL and BLB signal lines. Specifically, a pre-charge control circuit, described further below, provides a first control signal (BLEQB1) to the first pre-charge component **210** and a second control signal (BLEQB2) to the second pre-charge component **212**. A multi-stage pre-charge circuit, such as the one illustrated in FIG. 2 at **208**, can provide fast memory operation while avoiding anomalous memory operation.

(18) As noted above, to minimize pre-charge-related latency, it may be desirable to begin the next pre-charge cycle as quickly as possible after a read operation completes. FIG. 3 provides timing diagrams associated with exemplary operations in embodiments of the disclosure. In a first example at **302**, a pre-charge circuit is operated with reliance on only a single stage (e.g., the second pre-charge component **212** of FIG. 2). When operating in-specification, the control signal BLEQB for the single stage transitions from high to low levels to begin pre-charging as soon as possible following the conclusion of a read operation, whose duration is indicated by the high WL signal. Specifically, during the read operation that starts when WL transitions high, one of BL/BLB is pulled low by the specific T/C node that holds a low logic level. That pulling low of one of BL/BLB is illustrated by the downward sloping BL/BLB signal. At a point during the read operation (e.g., near the end of the period where WL is high), a sense amplifier senses the difference between BL and BLB, caused by one of those signals being pulled low, where the sense amplifier outputs a corresponding sensed data value for the bit cell. The BL/BLB signal that was pulled low must then be returned to a high level (i.e., pre-charged) to facilitate a next read operation. Some implementations aim to start pre-charging by bringing BLEQB low as soon after the read operation is over, when WL goes low, to ready the bit cell for a next read operation, where both BL and BLB are at a high level.

(19) But in reality, signal timing in a memory may not operate exactly as designed. Process variations, parasitic capacitances, or other factors may result in signals being received at a different time than designed. An example of such anomalous signal operation is illustrated at **304**. There, the pre-charge control signal BLEQB that was intended to transition low just after WL goes low, actually transitions low before WL goes low at the end of the read operation. This results in the one of BL/BLB that was being pulled low by the read operation being recharged to a high level prematurely while WL is still high. The rising BL/BLB signal may result in coupling with the WL signal, illustrated at **306**, which may in some instances disturb the contents of the T/C nodes in the bit cell, as illustrated at **308**. There, the levels at the T and C nodes, and correspondingly the contents of the bit cell are changed during a read operation, an anomalous result.

(20) The example at **310** illustrates control of a multi-stage pre-charge circuit that in embodiments enables fast pre-charging while mitigating the risk of anomalous circuit behavior. Specifically, a first pre-charge component (e.g., FIG. 2, **210**) is controlled by a first control signal BLEQB1 and a second pre-charge component (e.g., FIG. 2, **212**) is controlled by a second control signal BLEQB2. In this example, the first control signal BLEQB1 activates the first pre-charge component at **312** prior to the conclusion of the read operation, before WL goes low. The one of BL/BLB that had been pulled low begins to pre-charge, but at a slower rate than the examples of **302**, **304**. At **314**, following WL going low at the end of the read operation, the second control signal BLEQB2 activates the second pre-charge component. This activation increases the rate of pre-charging of BL/BLB to a high level (e.g., where the first pre-charge component is configured to charge at a first average rate between **312** and **314**, and the second pre-charge component is configured to charge, alone or in combination with the first pre-charge component, at a second average rate after **314**, the

second average rate being faster than the first average rate). While some coupling between the BL/BLB signal line being charged and the WL signal may occur at **316**, with corresponding minor disturbances at the T/C nodes between **312** and **314**, those disturbances are not sufficient enough to disturb (e.g., flip) the bit cell data contents.

(21) FIG. **4** is a block diagram depicting a pre-charge module for controlling a first pre-charge component and a second pre-charge component of a pre-charge circuit in accordance with embodiments. A pre-charge control circuit **110** is responsive to a timing circuit **402** and an input signal **404** that indicates that a read operation is to occur (e.g., the WL signal itself or another indicator). The pre-charge control circuit **110** includes a first pre-charge controller **406** that outputs a first control signal (BLEQB1) for controlling a first pre-charge component and a second pre-charge controller **408** that outputs a second control signal (BLEQB2) for controlling a second pre-charge component. In one example, upon receiving a read indication signal **404**, both the first and second pre-charge controllers **406**, **408** ensure that their control signals BLEQB1, BLEQB2 are in states that inhibit pre-charging during some or all of the read operation. Based on a known time associated with a read operation and signals from the timing circuit (e.g., a clock circuit) **402**, the pre-charge control circuit **110** controls the start of pre-charging of the one of BL/BLB that is pulled low. In one example, the first pre-charge controller **406** is configured to enable a first pre-charge component via BLEQB1 shortly before the conclusion of the read operation (e.g., 90% of the way through the duration of the read operation before WL goes low). The second pre-charge controller **408** is configured to enable the second pre-charge component, which may operate alone or in conjunction with the first pre-charge component, via BLEQB2 just after the conclusion of the read operation (e.g., at 110% of the duration of the read operation). Because the BL/BLB line is partially pre-charged when the read operation is completed, the second control signal may be asserted later than single stage pre-charging implementations (e.g., not as close to the WL transition) to avoid anomalous operation, while still returning the BL/BLB line promptly to a high level.

(22) Pre-charge components of a pre-charge circuit may take a variety of forms. FIG. **5** depicts example pre-charge components in accordance with embodiments. In a first example **502**, a pre-charge component (e.g., FIG. **2**, **210**) takes the form of an NMOS transistor positioned between a source node and a node connected to both the bit line/bit line bar signal lines, one of which is being pre-charged. When the first control signal BLEQ1 is high, the pre-charge component is activated.

(23) In a second example **504**, the pre-charge component comprises two PMOS transistors connected in series between a source node and a node connected to both the bit line/bit line bar signal lines. The two transistors have gates controlled by the first control signal. When the first control signal BLEQB1 is low, the pre-charge component is activated.

(24) In a third example **506**, the first pre-charge component comprises two PMOS transistors connected in series between a source node and the bit line/bit line bar signal lines. A first of the two transistors has a gate controlled by the first control signal. A second of the two transistors has a gate controlled by a signal based on a signal level at a node connected to both the bit line/bit line bar signal lines. When the first control signal BLEQB1 and one of the BL/BLB signals are low, the pre-charge component is activated.

(25) In a fourth example **508**, the first pre-charge component comprises an PMOS transistor between a second control signal and a node connected to both the bit line/bit line bar signal lines, the transistor having a gate controlled by the first control signal. When the first control signal BLEQB1 is low and the second control signal is high (e.g., when the second control signal is inhibiting the second pre-charge component), the pre-charge component is activated.

(26) FIG. **6** is a diagram depicting another example pre-charge circuit in accordance with an embodiment. In the example of FIG. **6**, a pre-charge circuit includes a first pre-charge component **602** and a second pre-charge component **604**. The first pre-charge component **602** comprises a PMOS transistor between a first node **606** connected to the bit line signal line and a second node **608** connected to the bit line bar signal line, the transistor having a gate controlled by the first

control signal BLEQB. The second pre-charge component **604** comprises two second stage transistors having gates controlled by the second control signal BLPREB, one second stage transistor being positioned between a source node and the first node **606**, and another second stage transistor being positioned between the source node and the second node **608**. In one example of operating the pre-charge circuit of FIG. **6**, the control signals BLEQB, BLPREB are raised to high levels near (e.g., just before, at the same time, just after) the time that WL goes high to initiate a read operation. One of BL/BLB is pulled low over time according to the status of the T and C nodes. Before the read operation ends when WL goes low, the first control signal BLEQB is brought low to initiate pre-charging via the first pre-charge component **602**. The one of BL/BLB that was pulled to a low level begins to pre-charge high, with some corresponding pulling down of the other bit line via equalization, during the remainder of the read operation. This may result in some coupling and corresponding effect on the WL signal. But that coupling is insufficient to affect the status of the T/C nodes. After the conclusion of the read operation when WL goes low, the second control signal BLPREB is brought low to initiate pre-charging via the second pre-charge component **604** in combination with the first pre-charge component **602**, promptly bringing both BL and BLB to high states in preparation for a next read operation.

(27) As noted above, coupling between a pre-charging BL/BLB and the WL signal can result in anomalous bit cell behavior. In some embodiments, limiting a magnitude of that coupling is desirable. FIG. **7** is a diagram depicting operation of control signals to limit a magnitude of word line coupling in accordance with embodiments. In operating the pre-charge circuit of FIG. **6**, a first simulation at **702** illustrates operation of only the second pre-charge component **604** via the BLPREB signal activation just before the WL signal transitions low. In this simulation **702**, the BLB signal has been pulled low via the read operation and is re-charged at high speed when BLPREB goes low, resulting in a 124 mv disturbance of the WL line.

(28) The second **704** and third **706** simulations utilize both pre-charge components **602**, **604** of FIG. **6**. In the second simulation **704**, the first control signal BLEQB activates and initiates pre-charging of BLB and a corresponding pulling down of BL via equalization that occurs when the first pre-charge component is activated. This results in a 46 mv disturbance of the WL line, which limits the risk of anomalous bit cell behavior compared to the 124 mv disturbance at **702**. The BL/BLB signals are brought the remainder of the way high by activation of the second pre-charge component when BLPREB goes low, just before the WL goes low.

(29) In the third simulation **706**, the first control signal BLEQB (and first pre-charge component) is activated earlier than the second simulation **704**. This results in an even smaller disturbance to WL, 10 mv, illustrating that timing of activation of the first and second pre-charge components can be controlled to mitigate bit line to WL coupling to an acceptable level.

(30) A first pre-charge component may be positioned in a variety of locations within a memory circuit. FIG. **8** is a diagram depicting a memory circuit where a first pre-charge component also serves as a second pre-charge component for another memory bank in accordance with embodiments. Specifically, a BL of an upper memory bank BL_UP is connected with a BL of a lower memory bank BL_DN via a PMOS transistor controlled by a BNKEQB control signal. Similarly, a BLB of an upper memory bank BLB_UP is connected with a BLB of a lower memory bank BLB_DN via another PMOS transistor controlled by a BNKEQB control signal.

(31) During an upper bank read operation, a word line associated with the upper bank WL_UP goes high and a control signal for the pre-charge circuit of the upper bank BLEQB_UP and BNKEQB go high to inhibit pre-charging during the read operation, such that one of BL_UP/BLB_UP can be pulled low according to the corresponding data stored in the upper bank bit cell. Because a read operation is not occurring in the lower bank, both BL_DN and BLB_DN have been charged high. The connections to the upper bank BL/BLB via the two PMOS transistors controlled by BNKEQB enable BL_DN and BLB_DN to be used as a source for a first-stage pre-charge of BL_UP/BLB_UP at the conclusion of the upper bank read operation. Specifically, near the end of

the read operation (e.g., before WL_UP goes low), BNKEQB is brought low to enable some pre-charging of BL_UP/BLB_UP from the BL_DN/BLB_DN lines, such that the pre-charging circuit **804** of the lower bank operates as both a first pre-charge component for the upper bank and a second pre-charge component for the lower bank. At a later time, BLEQB_UP is brought low enabling the remainder of pre-charging of BL_UP/BLB_UP to occur.

(32) FIG. **9** is a flow diagram depicting a method of controlling a multi-stage pre-charge circuit for charging one of a bit line and bit line bar signal line associated with a bit cell prior to a read operation in accordance with embodiments. The method is described with reference to structures described above for ease in understanding, but the method is also applicable to many other structures as well. The method includes providing a first control signal (BLEQB1) to a first pre-charging circuit (**210**) inhibiting charging during at least a portion of a read operation at **902**, where a word line (WL) associated with the bit cell (**202**) is activated during the read operation. A second control signal (BLEQB2) is provided at **904** to a second pre-charging circuit (**212**) inhibiting charging during at least the portion of a read operation. The first control signal (BLEQB1) is transitioned at **906** to enable charging by the first pre-charging circuit (**210**) prior to deactivation of the word line (WL), and the second control signal (BLEQB2) is transitioned at **908** to enable charging by the second pre-charging circuit (**212**) after transitioning the first control signal (BLEQB1).

(33) According to some embodiments, a memory circuit that includes a bit cell responsive to a bit line signal line and a bit line bar signal line configured to store a bit of data. A pre-charge circuit is configured to charge one of the bit line and bit line bar signal lines prior to a read operation, where the pre-charge circuit includes a first pre-charge component and a second pre-charge component, the first and second pre-charge components being individually controllable for charging the bit line and bit line bar signal lines.

(34) In embodiments, a method of controlling a multi-stage pre-charge circuit for charging one of a bit line and bit line bar signal line associated with a bit cell prior to a read operation includes providing a first control signal to a first pre-charging circuit inhibiting charging during at least a portion of a read operation, where a word line associated with the bit cell is activated during the read operation. A second control signal is provided to a second pre-charging circuit inhibiting charging during at least the portion of a read operation. The first control signal is transitioned to enable charging by the first pre-charging circuit prior to deactivation of the word line, and the second control signal is transitioned to enable charging by the second pre-charging circuit after transitioning the first control signal.

(35) In certain embodiments, a memory includes a two-dimensional array of bit cells, each row of bit cells being associated with a data word, each column of bit cells being associated with a multi-stage pre-charge circuit. A plurality of multi-stage pre-charge circuits are responsive to a pre-charge control circuit configured to provide a first control signal to first pre-charge components of the pre-charge circuits and a second control signal to second pre-charge components of the pre-charge circuits.

(36) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A circuit comprising: a bit cell responsive to a bit line and a bit line bar; a pre-charge circuit including: a first pre-charge component configured to charge the bit line and the bit line bar prior to conclusion of a read operation; and a second pre-charge component configured to charge the bit line and the bit line bar at the end of the read operation, wherein: the first pre-charge component of the pre-charge circuit includes a first transistor and a second transistor; the first transistor has a gate terminal configured to receive a first control signal and a source/drain terminal connected to a first source/drain terminal of the second transistor; and the second transistor has a gate terminal and a second source/drain terminal connected to each other and to both the bit line and the bit line bar; and a pre-charge control circuit configured to provide the first control signal to the first pre-charge component and a second control signal to the second pre-charge component, wherein: the first control signal is transitioned from inhibiting charging to enable charging by the first pre-charging component prior to deactivation of a word line; and the second control signal is transitioned to from inhibiting charging to enable charging by the second pre-charge component after transitioning of the first control signal.
2. The circuit of claim 1, wherein the second pre-charge component includes a third transistor having a gate terminal configured to receive the second control signal.
3. The circuit of claim 2, wherein the second pre-charge component further includes a fourth transistor having a first source/drain terminal connected to a source/drain terminal of the third transistor, a gate terminal connected to the gate terminal of the third transistor, and a second source/drain terminal connected to the bit line.
4. The circuit claim 2, wherein the second pre-charge component further includes a fourth transistor having a first source/drain terminal connected to a source/drain terminal of the third transistor and a gate terminal and a second source/drain terminal connected to each other and the bit line.
5. The circuit of claim 2, wherein: the second pre-charge component comprises a fourth transistor having a source/drain terminal connected to the first source/drain terminal of the third transistor and a fifth transistor having a source/drain terminal connected to a second source/drain terminal of the third transistor.
6. The circuit of claim 2, wherein: the third transistor having a second source/drain terminal connected to the bit line bar.
7. The circuit of claim 1, wherein the pre-charge control circuit is configured to activate the first pre-charge component of the pre-charge circuit prior to a word line signal transition indicating a conclusion of the read operation.
8. The circuit of claim 7, wherein the pre-charge control circuit is further configured to activate the second pre-charge component of the pre-charge circuit after the word line signal transition.
9. The circuit of claim 1, wherein the pre-charge circuit is associated with a plurality of bit cells, each bit cell being associated with a different data word.
10. The circuit of claim 1, wherein the pre-charge circuit includes more than two pre-charge components.
11. The circuit of claim 1, wherein the second pre-charge component includes a third transistor having a gate terminal configured to receive the second control signal and a first source/drain terminal configured to receive the first control signal.
12. The circuit of claim 11, wherein the third transistor of the second pre-charge component further includes a second source/drain terminal connected to the bit line.
13. A method comprising: providing a first control signal to a first pre-charge component inhibiting charging of a bit line or a bit line bar; providing a second control signal to a second pre-charge component inhibiting charging of the bit line or the bit line bar, wherein the first pre-charge component includes a first transistor having a gate configured to receive the first control signal, a first source/drain terminal configured to receive the second control signal, and a second source/drain terminal connected to both the bit line and the bit line bar; and the second pre-charge

component includes a third transistor and a fourth transistor having a first source/drain terminal connected to a source/drain terminal of the third transistor, a gate terminal connected to a gate terminal of the third transistor, and a second source/drain terminal connected to the bit line; transitioning the first control signal to enable charging of the bit line or the bit line bar prior to deactivation of a word line; and transitioning the second control signal to enable charging of the bit line or the bit line bar after the deactivation of the word line.

14. The method of claim 13, wherein a pre-charge control circuit is configured to provide the first control signal to the first pre-charge component and the second control signal to the second pre-charge component.

15. The method of claim 13, wherein the first control signal is transitioned from inhibiting charging to enable charging by the first pre-charging component prior to deactivation of a word line.

16. The method of claim 15, wherein the second control signal is transitioned to from inhibiting charging to enable charging by the second pre-charge component after transitioning of the first control signal.

17. The method of claim 13, wherein the gate terminals of the third and fourth transistor of the second pre-charge component are configured to receive the second control signal.

18. The method of claim 13, wherein the first transistor of the first pre-charge component and the third and fourth transistors of the second pre-charge component are PMOS transistors.

19. A memory comprising: an array of bit cells; a pre-charge circuit associated with the bit cells; and a pre-charge control circuit configured to provide a first control signal to a first pre-charge component of the pre-charge circuit and a second control signal to a second pre-charge component of the pre-charge circuit, wherein the first pre-charge component is configured to charge a bit line or a bit line bar prior to conclusion of a read operation and the second pre-charge component is configured to charge the bit line or the bit line bar at the end of the read operation, wherein the first pre-charge component includes a first transistor and a second transistor, the first transistor has a source/drain terminal connected to a first source/drain terminal of the second transistor, the second transistor has a gate terminal and a second source/drain terminal connected to each other and to both the bit line and the bit line bar, the first transistor has a gate terminal configured to receive the first control signal; wherein: the first control signal is provided to the first pre-charge component inhibiting charging during a portion of a read operation; a word line associated with a bit cell is activated during the read operation; the second control signal is provided to the second pre-charge component inhibiting charging during at least the portion of a read operation; the first control signal is transitioned to enable charging by the first pre-charge component prior to deactivation of the word line; and the second control signal is transitioned to enable charging by the second pre-charge component after transitioning the first control signal.

20. The memory of claim 19, wherein the second pre-charge component includes a third transistor and a fourth transistor, gate terminals of the third and fourth transistors are configured to receive the second control signal.
